



Herbivores as predators: how an arboreal herbivore encounters and impacts declining bird populations

Bianca McBryde^{a,*}, Catherine J. Price^a, Robert Heinsohn^b, Peter B. Banks^a

^a School of Life and Environmental Sciences, The University of Sydney, New South Wales, 2006, Australia

^b Fenner School of Environment and Society, Australian National University, Canberra, 2601, Australia

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ABSTRACT

Predation by carnivores is fundamental to ecological systems, however, not all prey are consumed by classic predators. Increasing evidence shows that seemingly obligate herbivores may supplement their diet with vertebrate prey, yet little is known about how they encounter or impact prey. Here we study the predation habits of an arboreal herbivore, the Common Brushtail Possum (*Trichosaurus vulpecula*), that has been recorded depredating multiple bird species, including the critically endangered Regent Honeyeater (*Anthochaera phrygia*). We conducted artificial nest experiments to test if possums seek out nests using odour and whether low predation rates were due to incidental predation or predation by specific individuals. We then used population viability analysis (PVA) to investigate the impact of this predation on extinction risk for a small declining population. We found no evidence that possums seek out nests using odour or of individual specialisation. Rather, our results suggest possums are incidental predators of birds' eggs, with possums only consuming eggs at 10% of sites. Regardless, PVA modelling demonstrated that low rates of predation by native arboreal mammals can substantially increase extinction risk by raising the threshold population density at which Allee effects cause negative population growth from 600 to 1000. These results have significant implications for conservation globally. They demonstrate how natural rates of incidental predation can become problematic for small populations, providing critical insights for conservation managers into the role of herbivores as predators and revealing how incidental native predation can act synergistically with other agents of decline to drive prey to extinction.

1. Introduction

Predation is a fundamental ecological process. It can underpin trophic cascades (Paine, 1980), regulate prey numbers (Krebs et al., 1995), alter ecosystem function (Estes et al., 2011), and, in cases of small declining populations, has the potential to drive prey to extinction (Heinsohn et al., 2022; Sinclair et al., 1998). A predator can be defined as any animal that consumes another living animal (Barbosa and Castellanos, 2005). While predators of vertebrates are classically conceived as those species that are particularly suited for finding prey through biological and ecological adaptations (Barbosa and Castellanos, 2005), even seemingly obligate herbivores can engage in the consumption of vertebrate prey. Accounts of such predation have been historically rare, with most herbivores thought to supplement their plant-based diet with invertebrate prey, if any (Reuter et al., 2023). However, the recent widespread availability of motion-activated wildlife cameras has

revealed a much wider variety of species involved in predation events than previously thought (Stojanovic et al., 2014; Taylor et al., 2018; Vazquez et al., 2023; Windhoffer and Pierce, 2021). White-tailed deer (*Odocoileus virginianus*), a textbook herbivore that shapes forest dynamics, have been recorded depredating birds' eggs and fledglings in multiple locations around the world (Ellis-Felege et al., 2008; Pietz and Granfors, 2000; Vazquez et al., 2023). Other classic herbivores that have been recorded depredating vertebrates include Red deer (*Cervus elaphus*) (Furness, 1988; Vazquez and Amico, 2023), Fallow deer (*Dama dama*) (Vazquez and Amico, 2023), Shetland sheep (*Ovis aries*) (Furness, 1988), Cattle (*Bos taurus*) (Nack and Ribic, 2005), Brown Lemurs (*Lemur fulvus*) (Mizuta, 2002; Nakamura, 2004), Howler monkeys (*Alouatta* spp.) (Lima and Bicca-Marques, 2024), and Common Brushtail possums (*Trichosurus vulpecula*) (Crates et al., 2019; Garnett et al., 1999).

Despite these widespread occurrences, very little is known about the foraging behaviour of such atypical predators, only that prey are

* Corresponding author at: The University of Sydney, New South Wales, 2006, Australia.

E-mail address: bmcb7499@uni.sydney.edu.au (B. McBryde).

reported infrequently in their overall diet. Low rates of prey occurrence in a predator diet could arise from two different mechanisms. Firstly, it is possible that only some individuals in the population are predators of a particular prey species. It is known that predation can be learned such that some individuals come to specialise on individual prey (Estes et al., 2003; Pelech et al., 2010), and this behaviour could also apply to herbivores that may learn to supplement their diet with vertebrate prey to meet unsatisfied nutritional requirements (White, 2011). If true, we would expect these individually specialised predators to be actively seeking out prey using sensory cues. Alternatively, predation may be incidental, where a forager fortuitously captures unexpected prey while searching for another source of food, leading to low levels of predation overall (Vickery et al., 1992). An important aspect of incidental predation is that the consumption of the incidental prey item does not prompt the predator to actively search it out in future (Vickery et al., 1992). However, despite its infrequency, incidental predation does not necessarily imply weak predation and it may still have strong effects on prey dynamics (Schmidt et al., 2001). The potential for incidental predators to have significant impacts on prey is especially true when predators are at high density and is particularly problematic for small and/or declining prey populations where every predation event is significant (e.g., threatened species; Crates et al., 2017). Furthermore, how incidental predators respond functionally to changes in prey density is likely to be starkly different from that of individually specialised predators (Vickery et al., 1992), making it particularly important to determine which of these two mechanisms is responsible for low predation rates.

Incidental predators and individually specialised predators likely differ in their ability to both drive decline in prey numbers and interact with existing causes of decline, such as Allee effects (Gascoigne and Lipcius, 2004). An Allee effect, describes a positive relationship between fitness and population size or density, where small populations actually experience a decrease in growth rate with decreasing population size (Allee, 1931; Stephens et al., 1999). In populations subject to demographic Allee effects, total fitness is depressed at low abundance and, if the effect is strong enough, population growth rate will be negative below a certain population size (Stephens and Sutherland, 1999). This negative growth rate can create a critical threshold, known as the 'Allee threshold', below which populations crash to extinction (Berec et al., 2007; Gascoigne and Lipcius, 2004; Stephens and Sutherland, 1999). Prior theory has considered how different types of predator functional responses vary in their ability to create Allee effects (Gascoigne and Lipcius, 2004). However, here we consider how low rates of natural predation may interact with existing Allee effects to become an agent of exaggerated decline.

Native predators are not usually considered to have an exaggerated impact on native prey due to the fact that they are co-evolved and as such prey are not naïve (Errington, 1956; Salo et al., 2007). Yet, native predators are increasingly considered problematic for small and declining populations across the globe (Bamber et al., 2024; Kavanagh, 1988; McMahon et al., 2020; Sweitzer et al., 1997). Even low rates of predation by predominantly herbivorous native species have been cited as a significant cause for concern for some endangered species (Heinsohn et al., 2022; Taylor et al., 2018). However, how and when these low rates of natural predation become problematic remains unknown. It is possible that when other factors, such as habitat loss, have driven species to decline, the additional pressure of predation by native herbivores may cause demographic Allee effects to reach a critical threshold at larger population sizes than they would if this predation was not present in the system, causing the population to crash to extinction earlier. Understanding the potential interaction between low rates of native predation and Allee effects will be extremely important for conserving small and declining populations where every predation event matters.

Here we focus on the role of predominantly herbivorous native arboreal mammals as predators of the critically endangered Regent Honeyeater (*Anthochaera phrygia*). Regent Honeyeaters are brightly

coloured, nomadic songbirds that were once abundant and widespread across the southern states of Australia (Franklin et al., 1989), but now only 300 individuals are estimated to be left in the wild (Commonwealth of Australia, 2016; Crates, 2019). Historically, Regent Honeyeaters fed in large flocks and nested in aggregations. Thus, a drastic decline in numbers is believed to have been detrimental to their ability to survive and breed successfully (Ford et al., 1993; Heinsohn et al., 2022). As a result, populations suffer from significant demographic Allee effects, suppressing population growth and driving the species towards extinction (Crates et al., 2017; Heinsohn et al., 2022). Multi-million-dollar habitat restoration and captive breeding programs are being undertaken to conserve the species; however, these efforts are significantly hampered by high mortalities due to nest predation (Commonwealth of Australia, 2016; Taylor et al., 2018).

Several species of native arboreal mammals have been recorded unexpectedly depredating Regent Honeyeater nests, including the Common Brushtail Possum (*Trichosurus vulpecula*) (hereafter brushtail possum or brushtail) (Crates et al., 2019), as well as multiple glider species (Taylor et al., 2018). Brushtail possums are typically considered to be almost exclusively herbivorous in Australian systems (Hume, 1999; Freeland and Winter, 1975; Marsh et al., 2006; Monks and Efford, 2006; Scrivener et al., 2004; Wiggins et al., 2006). However, video-monitoring of nesting birds has revealed multiple instances of brushtails consuming eggs, nestlings, and even adult birds (Crates et al., 2019; Garnett et al., 1999; Moorhouse et al., 2003; Rogers et al., 2023). Despite these recordings, animal matter has rarely shown up in diet studies of the species (Herath et al., 2021; Sweetapple and Nugent, 1998) and when it does the information reported is limited (e.g., "unidentified hair" Cochrane et al., 2003; "eggshell" in one stomach Glen et al., 2012; "other" How and Hillcox, 2000). As such, it is not clear whether predatory behaviour is common among all brushtails or just certain individuals, nor do we know whether they actively seek out nests using sensory cues or happen upon them incidentally. There is evidence that brushtails use odour when foraging for plants (Mella et al., 2018). As such, we would predict that if brushtail possums regularly consume birds' eggs, then olfaction will be an important tool for increasing foraging efficiency when searching for nests, as with other mammalian predators (Hughes et al., 2010; Price and Banks, 2012). Knowing whether brushtails are attracted to bird nesting odours, and whether this behaviour is common across all individuals, will be a key step in assessing, and reducing, their impact on declining populations.

The first aim of this study was to investigate whether or not brushtail possums use odour cues to seek out birds' nests. If brushtails use odour to find birds' nests, we hypothesise that (a) nests with bird odour will be visited significantly earlier than nests without bird odour, and (b) brushtail possums will also spend significantly more time interacting with nests with bird odour than nests without bird odour.

Our second aim was to determine if low predation rates are due to individual specialisation or incidental predation. If predation by brushtails is performed by a few specialised individuals, we hypothesise that (a) nests with bird odour will be found and depredated quickly (0–2 nights) (Hughes et al., 2010), and (b) the same nest sites will be visited across both experiments. Alternatively, if brushtails are incidental predators of birds' nests, we hypothesise that (a) they will encounter nests with bird odour and nests without bird odour randomly and (b) time to first visit will be largely variable.

The third aim was to investigate how and when low rates of natural predation become unsustainable for small populations. We used Population Viability Analyses (PVA) to model the effect of different initial population sizes on the probability of extinction (PE) and stochastic population growth rate (r) of Regent Honeyeaters when predation by arboreal mammals was included in the system versus when it was excluded. We hypothesise that: (a) scenarios including nest failures caused by arboreal mammals would enter a greater threshold of extinction risk at larger population sizes than those without this additional predation pressure; (b) scenarios including nest failures caused by

arboreal mammals will experience negative population growth rate (r) at larger population sizes than those without.

2. Methods

This work was conducted under Scientific License no. SL101415 from the NSW Department of Planning, Industry and Environment. Animal Care and Ethics approval was granted by the The University of Sydney (2022/2088). Experiments were carried out under the Australian Code for the Care and Use of Animals for Scientific Purposes. Care was taken to minimise any disturbance to animals during camera trap set-up and monitoring.

2.1. Nest predation experiments

2.1.1. Study site

Nest predation experiments were conducted at North Head, Sydney, Australia (33.8°S, 151.2°E) which supports a large population of brushtail possums (Price and Banks, 2021) and native birds including the New Holland Honeyeater (*Phylidonyris novaehollandiae*), which constructs a nest of similar shape and size to the Regent Honeyeater (Wright et al., 2023). Research was conducted in Austral Autumn (April and May 2023) which overlaps with the nesting period of the New Holland Honeyeater and other locally present birds (Garnett et al., 1999; Pyke and Recher, 1986; Recher, 1977).

Two experiments were undertaken, each with 40 nest sites spaced at least 200 m apart for independence, based on the 0.6–1 ha core home range of brushtail possums (Harper, 2005; Wat et al., 2020). One artificial nest was placed at each site (Fig. 1). Artificial nests are useful for quantifying relative predation rates between experimental treatments (Rangen et al., 2000; Smith et al., 2016). Nests were randomly allocated to one of two treatment groups (bird odour present or absent, $n = 20$ per treatment). This sample size was chosen to ensure adequate replication to run a Cox's Proportional Hazards Test, including covariates

(Ogundimu et al., 2016). Bird nesting odours were simulated using quail (*Coturnix japonica*) odour (manure, feathers, and eggshell) freshly collected from captive quails, as per (Fulton, 2006; Luck, 2003; Piper and Catterall, 2004; Price and Banks, 2012).

The nests were tree mounted at ~1.5 m height, within the typical range of New Holland Honeyeaters nests (McFarland, 1986). Percent canopy cover and diameter at breast height (DBH) of the mount tree were recorded at each site. DBH was at least 15 cm to ensure trees were large enough to support the weight of an adult brushtail. All nests were handled using latex gloves to prevent scent contamination from humans between treatments (Smith et al., 2016).

Nest visitors were recorded using a Scoutguard camera (series SG2060-X) mounted on a stake 1.5 m opposite each nest. Cameras recorded a 90-second video when triggered with no interval between triggers. The cameras were set to only trigger between 16:00 and 8:00 when possums were likely to be active (Herbert and Lewis, 1999). Both experiments were run for a total of 14 days, replicating the average incubation period of the New Holland Honeyeater (Yuni and Rose, 2005).

2.1.2. Experiment 1: do possums use bird odour to find nests?

Nests allocated to the Odour treatment received 10 g of quail manure and 5 large quail feathers, approximating the natural rate of deposition from a pair of nesting birds (Price and Banks, 2012). Feathers were hidden underneath the manure. Odour was refreshed every 3 days to ensure it remained attractive to predators (Bytheway et al., 2013). Control treatment nests were left empty as a visual control but still visited every 3 days. Artificial nests were modelled off locally present New Holland Honeyeaters (65 mm diameter, 30 mm high). For details see (Matthews et al., 1999).

2.1.3. Experiment 2: testing individual specialisation

To test whether low visitation rates were due to individual specialisation or incidental predation, a real quail egg was added to treatment

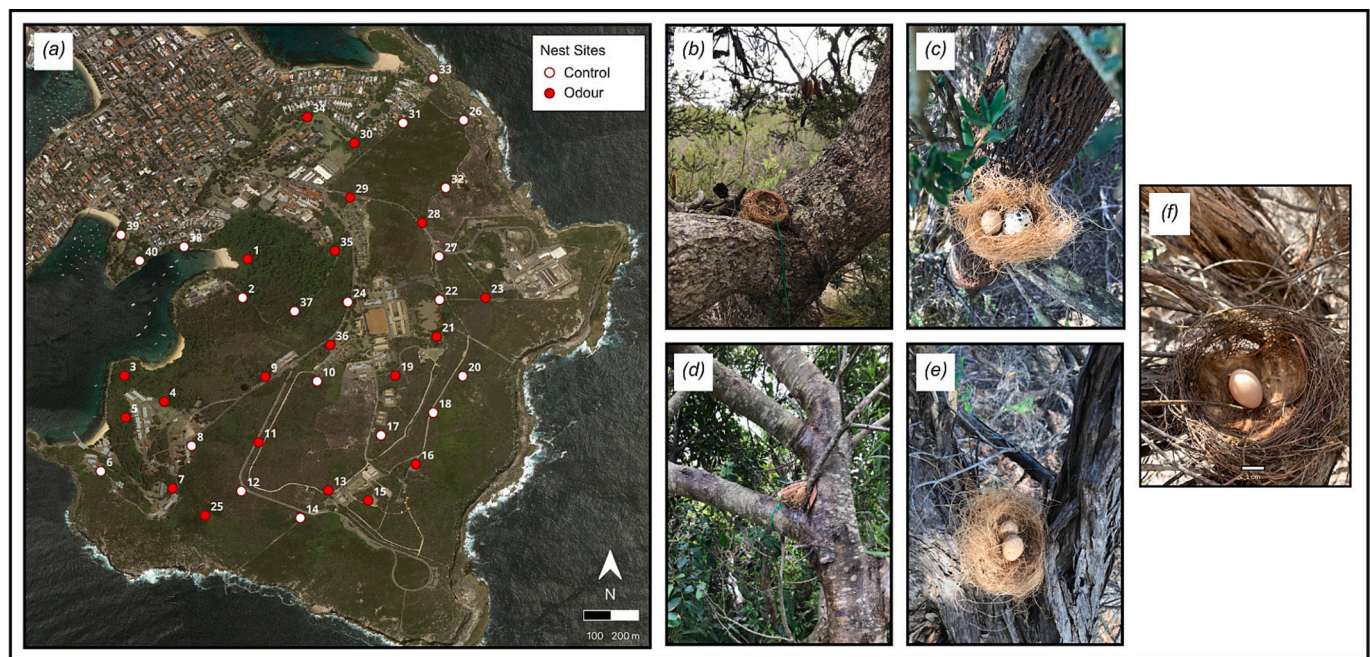


Fig. 1. Visual of experimental design. (a) Location of artificial nest sites at North Head, NSW, Australia. Map was constructed in QGIS (v.3.32). (b) Artificial nest at Odour site in Experiment 1, containing manure and feathers from quails (*Coturnix japonica*). (c) Artificial nest at Odour site in Experiment 2 containing 1 quail egg, 1 plasticine egg and 5 large feathers. (d) Empty artificial nest at Control site in Experiment 1. (e) Artificial nest at Control site in Experiment 2, containing two plasticine eggs. (f) Honeyeater spp. nest discovered at North Head by Emily Simpson (Australian Wildlife Conservancy). Nest discovered concurrently with the nest predation experiments conducted on site. Size and colour of the egg bears strong resemblance to (e). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

nests in the second experiment to simulate a prey reward. The Odour treatment included one quail egg (approximately 20 mm × 30 mm, cream coloured with brown and black speckles), one smaller plasticine egg (approximately 15 mm × 20 mm, cream coloured and unmarked; Rainbow modelling clay, New-bound P/L), and 5 large quail feathers freshly collected from nesting birds. We excluded manure in Experiment 2 in case possums did not associate it with nesting cues, despite past work indicating that other nest predators do (Price and Banks, 2012). Control nests included two plasticine eggs and no feathers. Plasticine eggs were used as a visual control and made using non-sulfurous plasticine moulded to match the eggs of the New Holland Honeyeater (Fig. 1). The same sites were used but new nests were mounted at least 5 m away from their previous position. Nests were crafted from a ball of coconut fibre shaped into a nest-like form to avoid any effects from Experiment 1 (Fig. 1). Nests were checked every 1–2 days to replace any eggs taken by non-target species. The Odour treatment was refreshed every 3 days if not disturbed, as in Experiment 1.

Camera trap data was reviewed by a single investigator (BM) blinded to treatments. Videos containing an animal had the following recorded: date, start time of the visit, species, number of individuals, total length of the visit, whether or not they interacted with the nest, and how long the interaction lasted. An *interaction* was defined as the head of an animal being in physical contact with the nest. Data from the second experiment included: if eggs were present at the start of a visit, and whether the eggs were disturbed or eaten. A *disturbance* was defined as notably moving or removing the egg. Eggs were recorded as *depredated* if the egg was seen to be consumed in the footage or substantial evidence of the egg being eaten was found the next morning (i.e., eggshell on the ground below the nest). The total number of nests visited, the time to first visit, and the time spent interacting with the nests was compared between treatments. Data from regular wildlife monitoring collected within a week of the experiments was used to confirm brushtail presence within the vicinity of each site.

2.1.4. Statistical analysis

Data visualisations were constructed in R Statistical Software (v4.2.1; R Core Team, 2022) via the package “ggplot2” (v3.3.3; Wickham, 2016). Low overall visit numbers prevented statistical analyses from being performed on number of visits, time to first visit, or length of interaction. To test whether mean percent canopy cover was significantly different between Odour and Control sites, a two-sample *t*-test was performed in base R (v4.2.1; R Core Team, 2022). Assumptions of normality and equal variance were confirmed using a Shapiro-Wilks test and an F-test. Homogeneity of variances was further confirmed using a boxplot.

2.2. Population viability analyses

We used the software package VORTEX 10 v.6.0 (Lacy and Pollak, 2023; software available at www.scti.tools/vortex) to investigate the effect of predation by arboreal mammals on the probability of extinction and population growth rate of Regent Honeyeaters at different initial population sizes. Vortex is an individual-based population simulation program that is appropriate for use with species that exhibit low fecundity and small population size and is particularly suited to modelling closed, single populations (Banks, 2004; Heinsohn et al., 2022; Lacy and Pollak, 2023; Miller and Lacy, 2005). The baseline population parameters were drawn from Heinsohn et al. (2022) (See Supplementary Methods Table S1). Sensitivity analyses were run on these parameters by Heinsohn et al. (2022). Adult and juvenile mortality were found to account for the highest proportion of variance (Heinsohn et al., 2022). A single (Blue Mountains) population was modelled as it is the largest remaining subpopulation and the dispersal rates between the three populations are extremely low (Crates, 2019).

2.2.1. Modelling predation rates

To simulate the effect of predation by arboreal mammals on the probability of extinction of Regent Honeyeaters, the PVA was run when predation by possums and gliders was included in the system versus when it was excluded. Predation by Sugar Gliders (*Petaurus breviceps*), and Squirrel Gliders (*Petaurus norfolcensis*) was included in the analysis as there is evidence that gliders are also opportunistic predators of birds' nests (Holland, 2001; Stojanovic et al., 2014). Estimates of predation rates by arboreal mammals were derived from three studies of honeyeater nest predation (Table 1). Each estimate has a varying degree of reliability due to either small sample size (Taylor et al., 2018), use of artificial nests (this study), or an inability to identify nest predators in 83% of predation events (Crates et al., 2019). To include all three estimates, we ran simulations under four different models (Table 2). MODEL 1 reflects the current reproductive success of Regent Honeyeaters in the wild under a broad demographic Allee effect, taking into account all causes of nest failure, including predation by arboreal mammals (Heinsohn et al., 2022). MODEL 2 uses the same parameters but increases female reproductive success by excluding arboreal mammal predation from the system, based on the most conservative estimate that arboreal mammals only contribute to ~4% of the predation on Regent Honeyeaters (LOW estimate; Crates et al., 2019). MODEL 3 also excludes arboreal mammal predation, however, it uses the estimate from this study that arboreal mammals contribute to ~17.5% of the predation (MID estimate; see Results). MODEL 4 similarly excludes predation by arboreal mammals, however, it is based on the higher estimate from video-monitoring of real Regent Honeyeater nests that arboreal mammals contribute to ~30% of the predation (HIGH estimate; Taylor et al., 2018).

Predation by arboreal mammals was assumed to follow a Type II functional response, where the number of nests depredated by arboreal mammals was assumed to be density dependent with no prey switching at low densities (Gascoigne and Lipcius, 2004). This models a scenario where predation by arboreal mammals is incidental based off the findings of the artificial nest predation experiments (Schmidt, 2004).

2.2.2. Effect of initial population size

To examine the effects of predation by arboreal mammals at different population sizes, 18 different scenarios were run for each model with the initial population size increased in increments of 50 individuals for each scenario until the historic population size of 1000 was reached (Heinsohn et al., 2022). Carrying capacity (K) was tested both increasing incrementally with initial population size and remaining at a historic level throughout the entirety of the modelling. These changes were found to have no effect on probability of extinction, so K was set at the historic rate of 3000 for all models.

2.2.3. Data analysis

The probability of extinction (PE) at each initial population size was extracted for all four models and visualised using a connected scatterplot constructed in R Statistical Software (v4.2.1; R Core Team, 2022) via the

Table 1

Estimates of predation by arboreal mammals on honeyeater nests taken from three studies.

Predation estimate	Source	Total depredated nests	Nests depredated by arboreal mammals	Contribution to predation (%)
LOW	Crates et al., 2019	51	2	3.9
MID	This study	23	4	17.5
HIGH	Taylor et al., 2018	10	3	30

Table 2

Models used in population viability analysis of the contemporary Blue Mountains population of Regent Honeyeaters (see Supplementary Table S1 for base population parameters).

Parameter	Model 1: All predation included	Model 2: LOW predation estimate excluded	Model 3: MID predation estimate excluded	Model 4: HIGH predation estimate excluded
Number of iterations	1000	1000	1000	1000
Number of years	20	20	20	20
Year length (days)	365	365	365	365
Predation	All predation	Excluding predation by arboreal mammals (LOW estimate)	Excluding predation by arboreal mammals (MID estimate)	Excluding predation by arboreal mammals (HIGH estimate)
Estimated predation by arboreal mammals	N/A	4% of total nest predation	17.5% of total nest predation	30% of total nest predation
Number of offspring produced per female per year (accounting for Allee effect) ^a	$= (0.85 + ((1.55 * (A = 1) + 2.55 * (A > 1))) * (N / 1400 + N))$	$= (0.89 + ((1.59 * (A = 1) + 2.59 * (A > 1))) * (N / 1400 + N))$	$= (1.025 + ((1.725 * (A = 1) + 2.725 * (A > 1))) * (N / 1400 + N))$	$= (1.15 + ((1.85 * (A = 1) + 2.85 * (A > 1))) * (N / 1400 + N))$
Initial population size	Increasing in increments of 50 from 150 to 1000	Increasing in increments of 50 from 150 to 1000	Increasing in increments of 50 from 150 to 1000	Increasing in increments of 50 from 150 to 1000
Carrying capacity (K)	3000 ^b	3000 ^b	3000 ^b	3000 ^b

^a See Supplementary methods Fig. S1 for explanation of how these formulae were derived. Allee effect is the Broad Allee effect outlined in Heinsohn et al. (2022). This was chosen based on advice from private communication with Rob Heinsohn.

^b Based on historic estimates of carry capacity as outlined in Heinsohn et al. (2022).

package “ggplot2” (v3. 3.3; Wickham, 2016). Mean stochastic population growth rate (r) for each scenario was assessed to determine at which initial population size r became negative for each model. A negative r indicates a population in decline and likely subject to demographic Allee effects (Berec et al., 2007; Stephens and Sutherland, 1999). We used the ICUN Red List criteria to assess when each model crossed into a new threshold of extinction risk. Specifically, we used the threshold of a 20% probability of extinction over 20 years, aligning with when a population moves from Vulnerable to Endangered (Mace and Lande, 1991).

3. Results

3.1. Nest predation experiments

Brush-tail possums were rarely recorded at nests over the 28 days of experimentation. Brush-tails visited only 20% of nest sites in each experiment and showed no distinct pattern of visitation between Odour and Control nests. A range of non-target avian and mammalian species were detected including the Australian Raven (*Corvus coronoides*), Black Rat (*Rattus rattus*), Brown Antechinus (*Antechinus stuartii*), Bush Rat (*Rattus fuscipes*), Common Ringtail Possum (*Pseudocheirus peregrinus*), Eastern Pygmy Possum (*Cercartetus nanus*), Noisy Miner (*Manorina melanocephala*), and Pied Currawong (*Strepera graculina*). Overall, animal activity was recorded at 39 out of the 40 nest sites. Canopy cover at nest sites fell within a range of 60 to 95% and mean canopy cover was

not significantly different between Odour and Control sites ($t = -0.7625$, $df = 38$, $p = 0.4055$).

3.1.1. Experiment 1

After 14 nights, 35 of the nest sites had been visited by avian or mammalian species (16 Odour, 19 Control). Brush-tail possums were recorded at eight nests (1 Odour, 7 Control) and six of the nests were visited by brush-tails more than once (Fig. 2). Time to first visit was highly variable across nests, ranging from 0 to 10 nights, and low visitation rate prevented comparisons across treatments. Brush-tails directly interacted with both Control and Odour nests during visits (Fig. 2). The time spent interacting was highly variable, ranging from 3 to 80 s across the visits.

3.1.2. Experiment 2

After 14 nights, 35 of the 40 nest sites had again been visited by avian or mammalian species (18 Odour, 17 Control), however, the sites visited varied from the previous experiment. Brush-tail possums were again recorded at eight nests (3 Odour, 5 Control) (Fig. 2). However, only two of the sites visited were the same as in the first experiment. Brush-tails ate eggs from four nests (2 Odour, 2 Control) and the two Odour nests were not revisited by brush-tails after the quail egg was eaten, despite the eggs being replaced. No nests were depredated more than once by a brush-tail possum. The time to first visit ranged from 0 to 8 nights.

Immediately following the nest experiments, 40 brush-tail possums were trapped on a single trap night during annual monitoring activities and 29 of the transects where brush-tails were caught overlapped directly with the nest sites (See Supplementary Results Fig. S3). Only 3 of the 40 nest sites were located greater than 100 m from a brush-tail capture.

3.2. Population viability analyses

Probability of extinction (PE) remains similar for all models at initial population sizes of 900 or more. The Blue Mountains population under current conditions (MODEL 1) breaches the 20% PE over 20 years criteria for listing as Endangered on the ICUN Red List at initial sizes of 650 or less (Fig. 3). When we remove the lowest estimate of predation by arboreal mammals (MODEL 2), populations cross this threshold at marginally smaller initial sizes of 600. However, when we remove the mid or high estimate of predation by possums and gliders (MODEL 3 and 4), populations don't meet the criteria for Endangered until their initial size drops as low as 400 or even 250 individuals (Fig. 3).

Populations remain in decline ($r < 0$) at initial population sizes of 1000 individuals (historic 1990s population estimate) for all models, except for MODEL 4 (Fig. 4). When the highest estimate of predation by arboreal mammals is removed from the system, a positive mean growth rate is achieved at initial populations greater than 600 individuals (Fig. 4). Simulated populations under these conditions do not cross the Allee threshold until they fall below an initial size of 600 individuals. For example population trajectories over time see Supplementary Results (Fig. S2).

4. Discussion

Our artificial nest predation experiments found no evidence that brush-tail possums were actively seeking out birds' nests, contrary to our first hypothesis. Rather, our results provide strong evidence for our alternative hypothesis that brush-tail possums as incidental predators of birds' eggs. Despite these results, population viability analysis found that the additional pressure of even incidental predation by arboreal mammals could increase the probability of extinction for small populations by up to four times. These results suggest the ecological significance of incidental predation by predominantly herbivorous species deserves more attention as a driver of population processes and conservation concern, particularly for threatened or declining species.

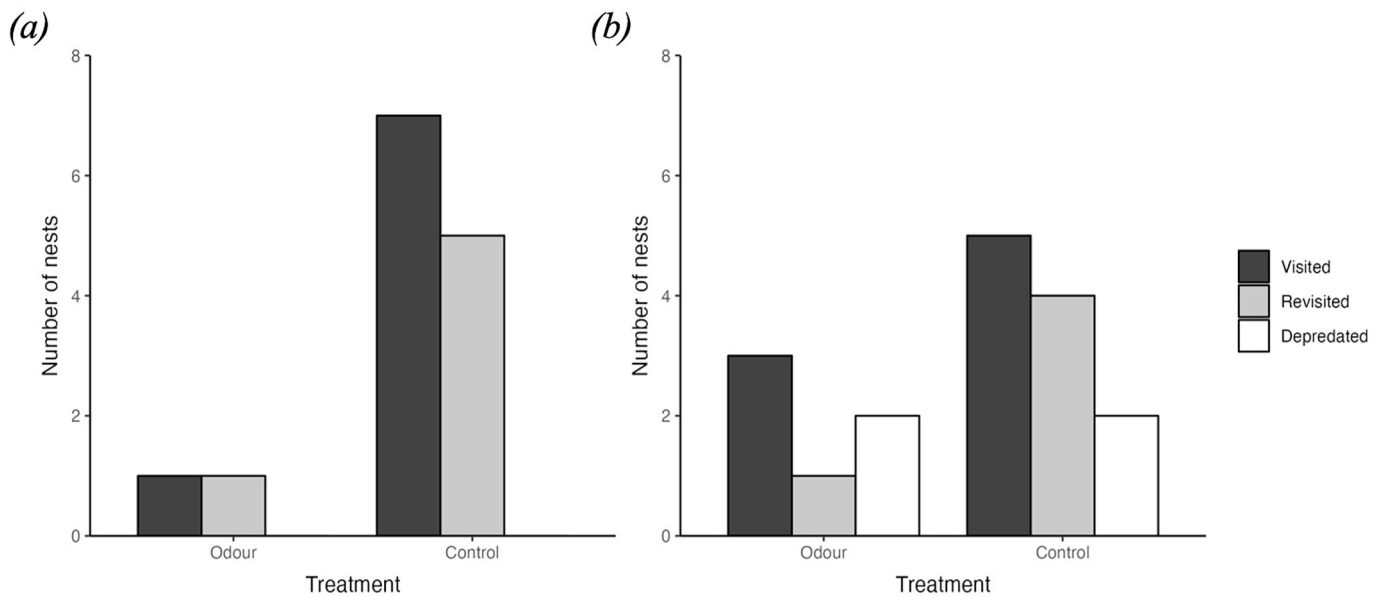


Fig. 2. Number of artificial nests visited or depredated by Common Brushtail Possums across two experiments manipulating nesting odours at North Head, NSW. (a) Total number of nests visited and revisited by Common Brushtail Possums over the 14 days of Experiment 1. Odour nests contained 10 g of quail manure and feathers from *Coturnix japonica* ($n = 20$) and Control nests were left empty ($n = 20$). (b) Total number of nests visited, revisited, and depredated by Common Brushtail Possums over the 14 days of Experiment 2. Odour nests contained 1 quail egg, 5 large feathers, and a plasticine egg ($n = 20$) and Control nests contained two plasticine eggs ($n = 20$). A nest was recorded as a *visited* if a Common brushtail possum triggered the camera at that site. A nest was recorded as *revisited* if a Common brushtail possum triggered the camera multiple times (at least 5 min apart) over the course of the 14-day experiment. A nest was recorded as *depredated* if either the quail egg or plasticine egg was seen to be consumed in the footage.

4.1. Brushtail possums as incidental predators

We found no evidence that herbivorous brushtail possums actively seek out birds' nests using odour cues. If brushtail possums were highly motivated to seek out and respond to the odour cues of open cup nesting birds, we expected to see brushtails visiting nest sites with bird odour significantly more and significantly earlier than nests without bird odour in both artificial nest predation experiments. However, neither of these patterns were reflected in the data, and fewer than 20% of the odour sites were visited by possums after 28 days of experimentation despite at least 40 individuals being confirmed in close proximity to nest sites.

Both specialist and generalist mammalian carnivores often use odour to locate prey from a distance, especially at night, and both can be rapidly attracted to prey scents (Hughes et al., 2010). Cats, foxes, and snakes visit plots scented with prey odour within just two days, while only a single control plot was visited in the same time period (Hughes et al., 2010). Black rats and other nest predators can easily discriminate at a distance between nests or bedding with prey odour and those without (Price and Banks, 2012; Ylönen et al., 2003). In contrast brushtail possums showed no evidence that they are attracted to sites with prey odour over sites without prey odour. Only one nest with quail manure, feathers, and eggshells was visited by brushtails after 14 days of odour treatment, despite this same cue being known to rapidly attract a range of other nest predators (Fulton, 2006; Luck, 2003; Piper and Catterall, 2004; Price and Banks, 2012; Smith et al., 2016). Furthermore, even when real prey (quail eggs) was added to the nest, brushtails were only recorded at three nests with bird odour over another 14 days.

If the low visitation rate recorded in this study was the result of only a few individuals in the population specialising in nest predation, we expected to see patchiness in nest visitation with the same nest sites visited across both Experiment 1 and 2. We also expected nests to be visited quickly (i.e., within 0–2 nights) as with other specialised mammalian predators (Hughes et al., 2010; Price et al., 2025). However, neither of these trends were observed. Time to first visit varied widely from 0 to 10 nights across both experiments. Furthermore, no nests with

eggs were depredated more than once by brushtail possums. If predation was learned, we expected to see brushtails returning to nest sites more quickly after receiving a food reward, as has been observed in other nest predation studies involving individually specialised predators (Pelech et al., 2010; Price et al., 2025). Rather, we observed no increase in the motivation of brushtail possums to pursue bird odour cues after consuming prey. This absence of behavioural change is a key diagnostic factor for identifying incidental predators (Vickery et al., 1992).

Our results suggest that past observations of predation events by brushtail possums reflect them opportunistically consuming eggs and nestlings, likely while foraging for blossoms and nectar. Blossoms are a primary food source for both brushtail possums and nesting honeyeaters (Crates et al., 2017; Smith and Murray, 2003), providing the key spatial and temporal coincidence of a primary food source and incidental prey that is required for incidental predation to occur (Schmidt et al., 2001). Mammalian herbivores are predicted to preferentially consume flowers, fruits, and seeds to meet protein and nutrient requirements that are difficult to obtain from a leaf-only diet (White, 2011). Supplementing their diet with protein-rich animal tissues such as birds' eggs may help fill this same nutritional gap (Kerle, 2001; Nugent et al., 2000; White, 2011). Given the lack of evidence that brushtail possums actively seek out these food sources, it is possible that only an occasional incidental predation event is sufficient to fulfil these protein requirements. However, even the occasional consumption of a prey item by many individuals may still have a large cumulative effect on prey populations (Schmidt et al., 2001). Additionally, while multiple locally present honeyeaters were actively breeding during this study, further research should be conducted during brushtail possum breeding seasons to determine if brushtails seasonally cue into bird nesting odours as a means of meeting minimum amino acid requirements for reproduction (White, 2011). It is also important to note using artificial nests to draw conclusions about overall predator behaviour must be done cautiously as they lack the auditory and visual cues of nestlings and adult birds. Further investigation into brushtail possum responses to auditory nesting cues would therefore be advisable to confirm the findings of this study.

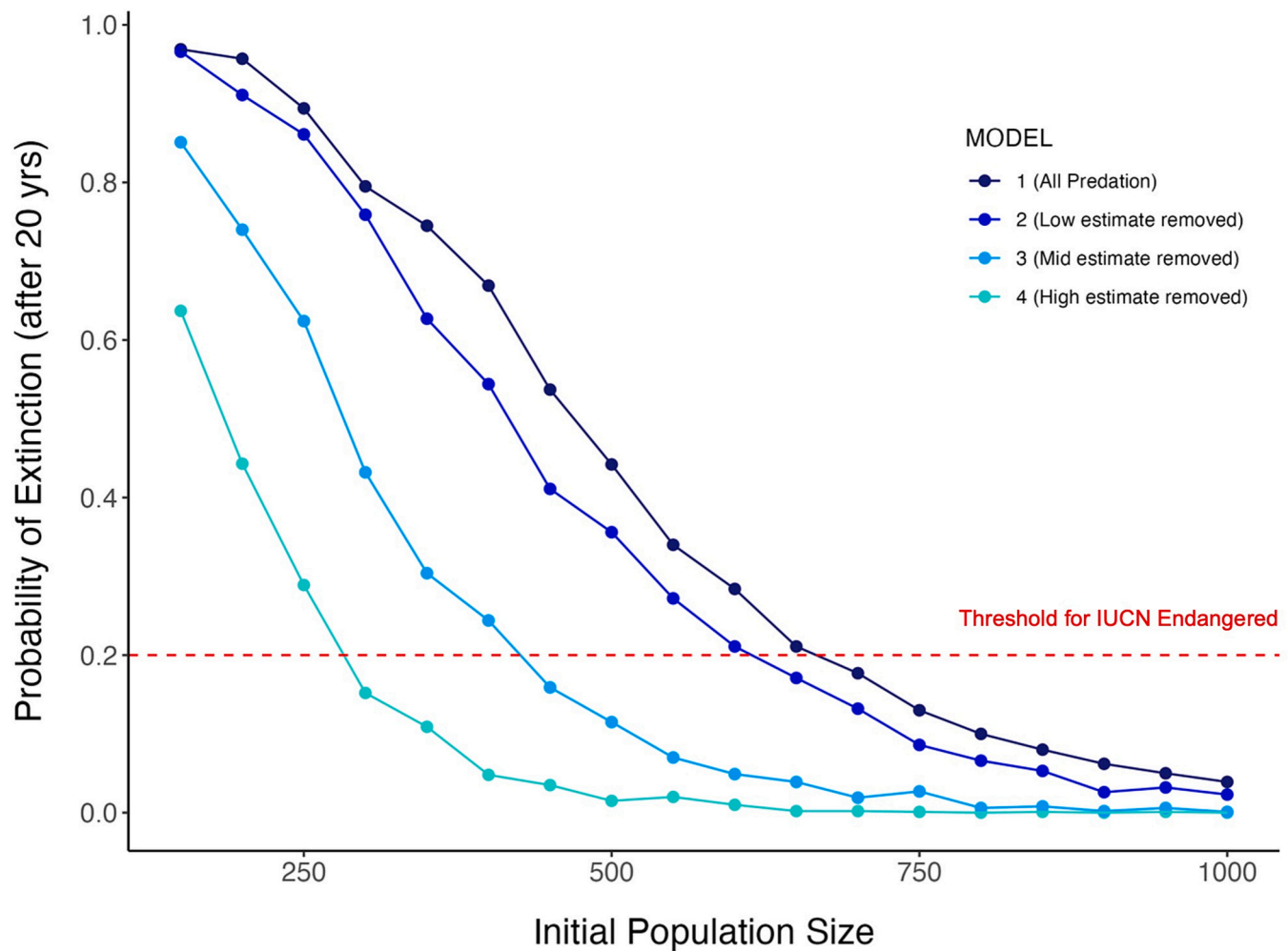


Fig. 3. Probability of extinction (PE) of the Regent Honeyeater at different population sizes when predation by arboreal mammals is included in the model versus when it is excluded. PE values represent the proportion of 1000 simulated populations that go extinct after 20 years. The red dotted line represents a 20% probability of extinction after 20 years. For detailed descriptions of the Models see Table 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.2. Effect of arboreal mammal predation on probability of extinction

Our simulations showed that even low rates of predation recorded by predominantly herbivorous arboreal mammals can have a notable effect on the persistence of small populations. At large population sizes (>900), predation by arboreal mammals appears to have no notable impact on the probability of extinction of Regent Honeyeaters. This result suggests that for historic population sizes, incidental predation by arboreal mammals is unlikely to have an exaggerated impact on the population. However, as the population decreases, the amount of predation by arboreal mammals in the system can have a detrimental impact on the survival of Regent Honeyeaters in the wild. Without predation by arboreal mammals, Regent Honeyeater populations do not cross the extinction risk threshold for the ICUN Red List “Endangered” category until they reach an initial population size as small as 300 individuals. However, when nesting failures caused by arboreal mammals are included in the model, they cross this threshold at populations of more than double this size. Furthermore, under current conditions, Regent Honeyeaters would be in decline (mean growth rate < 0) even at large initial population sizes of 1000 individuals. However, when we remove the highest estimated rate of arboreal mammal predation from the system, populations do not cross the ‘Allee threshold’ and enter decline until they reach a starting population of less than 600 individuals. This finding provides evidence for an interaction between

native incidental predation and Allee effects, suggesting that low rates of predation can exacerbate Allee effects to cause populations to enter a state of negative growth rate earlier than they would without the nest failures caused by incidental native predators.

The modelling in this paper demonstrates, unsurprisingly, that the impact of incidental predators on prey increases as their estimated predation rates increase. Further study into nest predation by arboreal mammals will therefore be essential for determining which of the estimates in this study provides the best assessment of the impact of these predators on small populations. Additionally, while the PVA accurately models the predicted behaviour of incidental predators at low prey densities following a Type I or Type II functional response, the assumption of a positive linear relationship between prey density and predation rate likely led to an overestimation of the impact of predation at higher prey densities. Future studies could therefore incorporate a functional response curve into population models to better account for predator consumption rates plateauing as prey densities increase beyond a certain threshold.

4.3. Broader implications

In this study, we demonstrate that incidental native predators can interact with other agents of decline, forming synergistic impacts on native prey (i.e., an effect greater than the sum of their independent

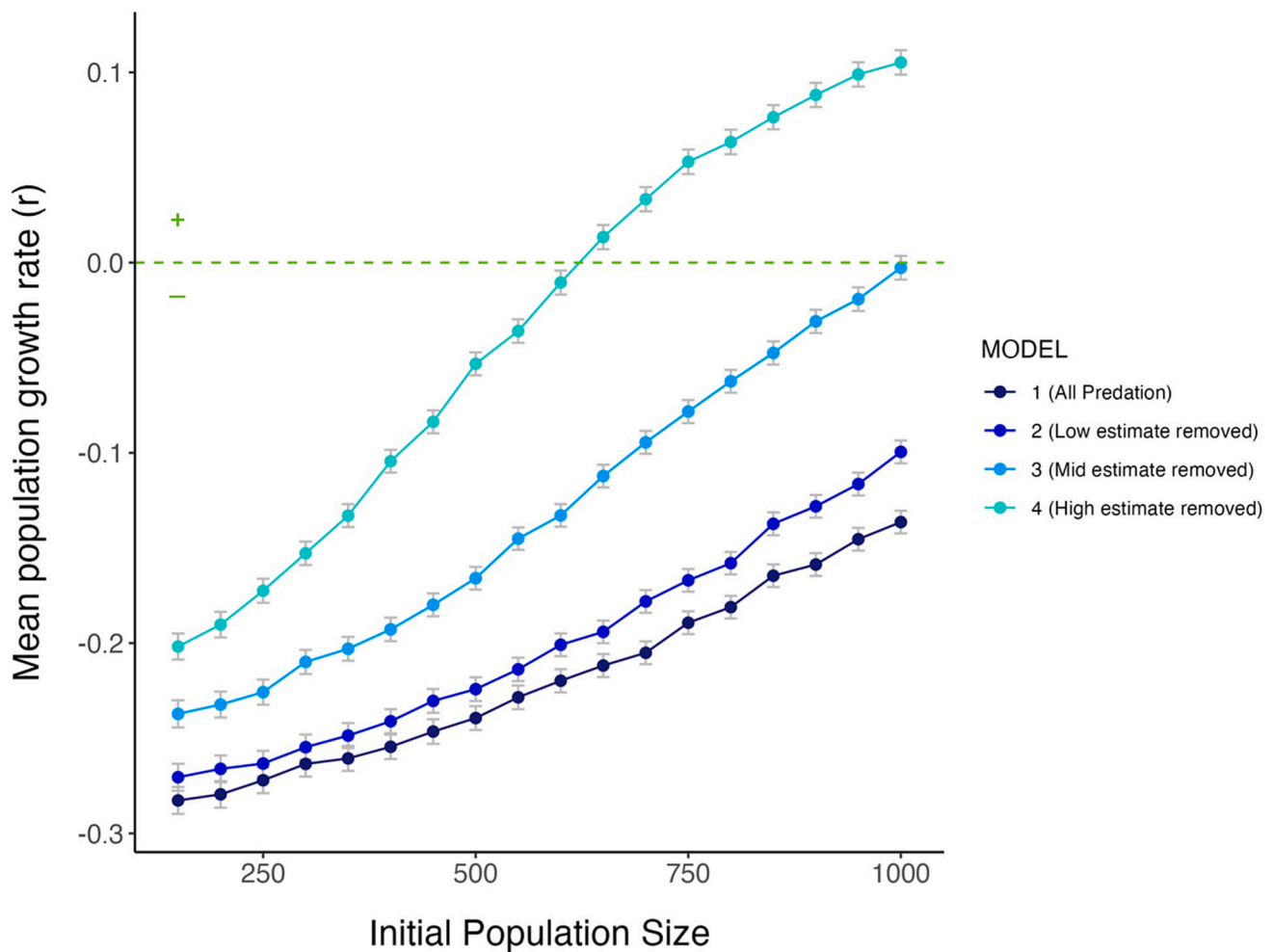


Fig. 4. Mean simulated growth rates for populations of Regent Honeyeaters under different levels of predation and different initial population sizes. (r) represents the mean stochastic growth rate for populations of Regent Honeyeaters modelled in VORTEX 10 v.6.0 (Lacy and Pollak, 2023) over a 20-year period. Error bars represent standard error (SE). Dotted green line represents $r = 0$. For explanation of the Models see Table 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

impacts; Doherty et al., 2015). There is a strong emerging area of research on the synergistic impacts of invasive predators and other ecological disturbances such as fire and habitat degradation on native mammals (Hradsky, 2020; Doherty et al., 2015). There is also a growing body of evidence demonstrating a relationship between habitat loss and increased predation pressure by native carnivores in Europe (Roos et al., 2018). However, little to no research has been done on how these agents of decline might interact with natural systems to form synergies with incidental native predators. This study provides one of the first quantitative illustrations of how such an interaction might occur with low rates of predation by native species contributing to driving native prey to extinction.

Incidental predation by native herbivores is likely sustainable for larger, more robust prey populations as the species in question are co-evolved and have a history of co-existing. However, disturbing our natural environments through habitat degradation, introduction of non-native species, and climatic changes means that a previously acceptable level of harvesting by incidental native predators can become extremely problematic. As our ecosystems change, interactions between species that would not usually require intervention are emerging as critical conservation challenges (Hobbs et al., 2009). Managing these interactions is particularly complex as it pits the protection of one native species against another.

The potential for occasional predation by herbivores to have significant impacts on small and declining prey populations has wide-ranging

implications for conservation globally. Numerous seemingly obligate herbivores have been recorded predating on vertebrate prey around the world (Bicca-Marques et al., 2009; Furness, 1988; Lima and Bicca-Marques, 2024; Nack and Ribic, 2005; Vazquez et al., 2023). These species have likely been overlooked as potential predators of threatened species in conservation programs, however, here we demonstrate that even occasional predation by herbivores can contribute substantially to the probability of extinction of endangered prey. Our findings suggest that policy makers and conservation managers may need to consider a much wider suite of potential predators when attempting to recover small populations, as our results suggest that failing to account for predation by herbivores could have dire impacts for the conservation of critically endangered species. Further research will be needed to determine when and how to intervene to reduce predation by herbivores in different ecological systems. Such interventions will be particularly difficult when herbivores are native to the system, as the removal of native predators to protect threatened prey raises significant ethical and ecological concerns (Woinarski, 2019). Behaviourally mediated non-lethal methods may be appropriate to manage these interactions (Bamber et al., 2024; Selonen et al., 2022), however, more research will be needed to determine their suitability for incidental predators. In a time of rapid ecosystem change, this potential battle between native species presents a wicked problem for the future of ecology.

CRediT authorship contribution statement

Bianca McBryde: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Catherine J. Price:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization. **Robert Heinsohn:** Writing – review & editing, Methodology. **Peter B. Banks:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The work is all original research carried out by the authors. All authors agree with the contents of the manuscript and its submission to the journal.

No part of the research has been published in any form elsewhere or is being considered for publication elsewhere.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biocon.2026.111855>.

Data availability

Data will be made available on request.

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